Reference Sheet

Math

```
# import math library to use math functions
import math

result = math.ceil(6.1)  # result == 7
result = math.floor(6.1) # result == 6
result = abs( -7)         # result == 7
result = math.sqrt(9)    # result == 3.0

# trigonometry
theta = math.pi           # theta == pi
result = math.sin(theta)  # result == 0.0
result = math.tan(theta)  # result == 0.0
result = math.cos(theta)  # result == -1.0
```

Arithmetic Operators

```
# add, subtract, multply, divide, exponentiate
result = 20 + 25   # result == 45
result = 20 - 25   # result == -5
result = 20 / 25   # result == 0.8
result = 4 / 2     # result == 2.0
result = 20 * 25   # result == 500
result = 2 ** 4    # result == 16
result = "a" + "b" # result == "ab"

# integer division
result = 20 // 25   # result == 0
result = 4 // 2     # result == 2
result = 4.0 // 2   # result == 2.0
result = 4 // 2.0   # result == 2.0

# modulus - - returns the remainder after division
result = 20 % 2   # result == 0
result = 25 % 2   # result == 1
result = 25 % 20  # result == 5
```

Comparisons and Logical Operators

```
# Equal to `==`, not equal to `!=`
result = 20 == 25   # result == False
result = 20 != 25   # result == True

# Less than `<`, greater than `>`,
result = 20 > 25   # result == False
result = 20 < 25   # result == True
result = 20 < 20   # result == False

# less than or equal to `<=`
# greater than or equal to `>=`
result = 20 <= 20   # result == True

# combining booleans using 'and'
result = 20 == 25 and 20 > 25  # result == False
result = 20 != 25 and 20 < 25  # result == True

# combining booleans using 'or'
result = 20 == 25 or 20 > 25  # result == False
result = 20 != 25 or 20 > 25  # result == True

# something is `in` a list, something is `not in` a list
result = 25 in [20, 25]       # result == True
result = 25 in [20, 26]       # result == False
result = 25 not in [20, 26]   # result == True

# Something is None:
result = "" is None         # result == True
result = 2025 is not None   # result == True
```

Type Conversions

```
result = int(4.2)           # result == 4
result = float("4.2")       # result == 4.2
result = str(4.2)           # result == "4.2"
result = bool(1)            # result == True
result = bool(0)            # result == False
result = int(True)          # result == 1
result = int(False)         # result == 0
```

Looping

```
# i goes from 0 .. 9
for i in range(10):
# i goes from 3 .. 9
for i in range(3, 10):
# i gets the value 3, then 5, then 7, then 9
for i in range(3, 10, 2):
# fruit gets the value "apple", then "banana", then "orange"
for fruit in ["apple", "banana", "orange"]:

# fruit gets the value "apple", then "banana", then "orange"
# index gets the value 0, then 1, then 2
for index, fruit in enumerate(["apple", "banana", "orange"]):

# zip allows you to go through multiple lists simultaneously
for a, b in zip(a_list, b_list):
for a, b, c in zip(a_list, b_list, c_list):

# break to exit a for loop early
for some_var in some_list:
  if some_condition:
    break
```

Functions

```
# An example function definition:
def foo(a, b):     # function 'foo' takes two parameters 'a' and 'b'
    return a + b   # returns the result of adding 'a' and 'b'

# Using the example function:
result = foo(1, 2)      # result gets the value 3
result = foo("1", "2")  # result gets the value "12"
```

Lists

```
# Create an empty list:
my_list = []
# Create a list with values:
my_list = ["apple", "banana", "orange"]


# Access a specific element in a list using an index:
result = my_list[0]         # result == "apple"
my_list[0] = "pear"         # my_list == ["pear", "banana", "orange"]

# Use negative index to get value in list in reverse order:
result = my_list[ -1]        # result == "orange"

# Get the length of a list:
result = len(my_list)       # result == 3

# Add something to the end of a list:
my_list.append("lime")
result = len(my_list)       # result == 4

# Get the maximum / minimum / sum of a list's items:
number_list = [1, 2, 3]
result = max(number_list)   # result == 3
result = min(number_list)   # result == 1
result = sum(number_list)   # result == 6

# Remove something from the end of a list:
result = number_list.pop()  # result == 3, number_list == [1, 2]

# And then remove something from a given index of a list:
result = number_list.pop(0) # result == 1, number_list == [2]

# Sort/reverse a list:
number_list = [3, 2, 1]
number_list.sort()       # number_list == [1, 2, 3]
number_list.reverse()    # number_list == [3, 2, 1]
```